

ANNIE STEVENSON

BIOINFORMATICIAN

WAYNE, PA 19087

610-731-6857 | astevenson1318@gmail.com | [linkedin.com/in/annie-stevenson13](https://www.linkedin.com/in/annie-stevenson13)

PROFESSIONAL SUMMARY

Senior research associate and aspiring bioinformatician experienced in microbiology, virology, and antimicrobial drug discovery. Skilled in bridging wet-lab expertise with computational biology. Hands-on experience automating pipeline sections for sequence analysis and data reporting. Adept at presenting scientific findings to cross-functional teams and external collaborators. Currently transitioning into a bioinformatics-focused role with a strong foundation in experimental biology, statistical analysis, and computational genomics.

SKILLS

- **Bioinformatics and Genomics** – Whole genome sequencing (WGS), SNP calling, genome assembly, MLST, protein mutation analysis, Geneious Prime, NCBI tools
- **Programming and Data Analysis** – Python, R, Bash, data visualization, reproducible pipelines
- **Molecular Biology and Microbiology** – Antimicrobial susceptibility testing, time kill assays, cytotoxicity assays, PCR and qPCR, ELISA, cell culture (insect, mammalian, bacterial), lentiviral assays
- **Collaboration and Communication** – Technical reporting, cross-team collaboration, conference presentations

WORK EXPERIENCE

VENATORX PHARMACEUTICALS, Malvern, PA

11/2020 – 07/2025

Biopharmaceutical company focused on discovering and developing novel therapies for multidrug-resistant bacterial infections.

Research Associate II, Microbiology (03/2024 – 07/2025)

Conducted bioinformatics-driven analyses of clinically relevant bacterial genomes to support drug discovery efforts and understand resistance mechanisms. Developed and applied computational workflows for WGS analysis, mutation identification, and sequence annotation.

- Coded Python script to detect and report protein mutations from MSAs, streamlining the WGS workflow and improving analysis speed.
- Conducted WGS analysis of *A. baumannii*, *P. aeruginosa*, and *S. maltophilia* strains, including read trimming, assembly, MLST, and SNP calling.
- Authored and presented a poster on novel Phase 1 drug combination ceftibuten-ledaborbactam at the ASM conference, contributing to the organization's scientific visibility.

Research Associate, Microbiology (11/2020 – 03/2024)

Executed microbiological assays and molecular biology workflows to characterize novel antimicrobial compounds. Supported regulatory filings and contributed to assay development and process improvements.

- Performed antimicrobial susceptibility testing and profiling of novel inhibitors according to CLSI guidelines.
- Mapped primers and probes to genes of interest to support RT-PCR assay development using Geneious Prime.
- Contributed to NDA submission documents for novel BL/BLI combination cefepime-taniborbactam.

WUXI ATU, Philadelphia, PA

09/2018 – 11/2020

Contract research organization specializing in cell and gene therapy development. Worked in the Advanced Therapies unit.

Virology Lab Technician III (04/2020 – 11/2020)

Contributed to assay development and execution in virology R&D projects, with a focus on viral stability and replication competence. Trained technicians and oversaw regulated workflows to ensure accuracy and compliance.

- Designed experiments for research, assessing the impact of temperature excursions on viral titers.
- Trained staff on replication-competent lentiviral assays, including ELISA, qPCR, and cell culture endpoints.

- Managed the replication-competent lentiviral (RCL) work cell, overseeing equipment, documentation review, and assay quality.

Virology Lab Technician II (09/2019 – 04/2020)

Supported assay transfers, workflow optimization, and regulated assay execution to ensure high-quality virology testing.

- Trained technicians on GMP-compliant assay execution and documentation.
- Advised on workflow and equipment needs during assay transfer to a new facility.
- Partnered with study directors to prepare and review methods for complex co-culture assays.

Virology Lab Technician I (09/2018 – 09/2019)

Performed regulated virology assays in support of therapeutic development programs. Maintained compliance with BSL2+ standards while generating high-quality data for client studies.

- Executed virology assays under GMP conditions for multiple therapeutic candidates.
- Supported laboratory operations and documentation review to ensure accuracy and reproducibility.

EDUCATION

Master of Science (M.Sc.), Bioinformatics, BRANDEIS UNIVERSITY, Boston, MA (Expected 2/2026)

Bachelor of Science (B.Sc.), Biology, MUHLENBERG COLLEGE, Allentown, PA

CERTIFICATIONS

Basic Statistics, COURSERA (2024)

Whole Genome Sequencing of Bacterial Genomes, COURSERA (2022)

MEMBERSHIPS

Member, INTERNATIONAL SOCIETY FOR COMPUTATIONAL BIOLOGY (2024 – Present)

PUBLICATIONS

Uehara T, Chatwin CL, Miller B, Edwards M, **Stevenson A**, Colombo J, Six DA, Daigle DM, Moeck G, Boyd SA, Pevear DC. Spectrum of cefepime-taniborbactam coverage against 190 β -lactamases defined in engineered isogenic Escherichia coli strains. Antimicrob Agents Chemother. 2025 May 7;69(5):e0169924. doi: 10.1128/aac.01699-24. Epub 2025 Apr 1. PMID: 40167371; PMCID: [PMC12057355.nt](#).

Uehara T, Chatwin CL, Miller B, Edwards M, **Stevenson A**, Colombo J, Uehara K, Daigle DM. Impact of porin deletions on cefepime-taniborbactam activity against Klebsiella pneumoniae. Antimicrob Agents Chemother. 2025 May 7;69(5):e0167224. doi: 10.1128/aac.01672-24. Epub 2025 Apr 2. PMID: 40172193; PMCID: PMC12073839.

Uehara T, Zulli AL, Miller B, Avery LM, Boyd SA, Chatwin CL, Chu GH, Drager AS, Edwards M, Emeigh Hart SG, Myers CL, Rongala G, **Stevenson A**, Uehara K, Yi F, Wang B, Liu Z, Wang M, Zhao Z, Zhou X, Zhao H, Stratton CM, Bala S, Davies C, Tkavc R, Jerse AE, Pevear DC, Burns CJ, Daigle DM, Condon SM. A new class of penicillin-binding protein inhibitors to address drug-resistant Neisseria gonorrhoeae. bioRxiv [Preprint]. 2024 Dec 27:2024.12.27.630553. doi: 10.1101/2024.12.27.630553. PMID: 39763734; PMCID: PMC11703265.

Myers CL, **Stevenson A**, Miller B, Daigle DM, Uehara T, Pevear DC. Cefepime-taniborbactam and ceftibuten-ledaborbactam maintain activity against KPC variants that lead to ceftazidime-avibactam resistance. Antimicrob Agents Chemother. 2025 Mar 5;69(3):e0151124. doi: 10.1128/aac.01511-24. Epub 2025 Feb 10. PMID: 39927783; PMCID: PMC11881561.

POSTERS

Stevenson A. K., Chatwin, C. L., Miller, B., Edwards, M., and Moeck, G. (2024, June 13 – 17). *In vitro* activity of ceftibuten-ledaborbactam and comparators against Enterobacterales from a Phase 3 study of adults with complicated urinary tract infection (cUTI) [Poster presentation]. ASM Microbe 2024, Atlanta, Georgia.

Koeth L., Fisher J., Belley A., Demetrio F., Chatwin C. L., Miller B., **Stevenson A.**, Edwards M., Cusick S., and Moeck G. (2025, April 11 – 15). Performance assessment of a cefepime-taniborbactam gradient diffusion strip for antimicrobial susceptibility testing of Enterobacterales and *Pseudomonas aeruginosa* [Poster presentation]. ESCMID Global 2025, Vienna, Austria.

Belley A., Hackel M., Wise M., Sahm D., Chatwin C.L., Miller B., **Stevenson A.K.**, Edwards M., Daigle D.M., Moeck G. (2025, April 11 – 15). Cefepime-taniborbactam (FTB) epidemiological cutoff values (ECOFFs) against key Enterobacterales species and *Pseudomonas aeruginosa* [Poster presentation]. ESCMID Global 2025, Vienna, Austria.